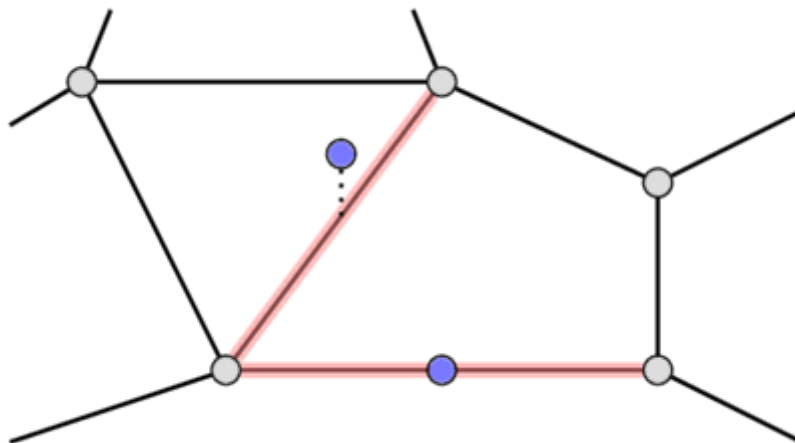


Point location in $O(\log n)$

Consider the following problem: you are given a **planar subdivision** without any vertices of degree one and zero, and a lot of queries. Each query is a point, for which we should determine the face of the subdivision it belongs to. We will answer each query in $O(\log n)$ offline. This problem may arise when you need to locate some points in a Voronoi diagram or in some simple polygon.

Algorithm

Firstly, for each query point $p(x_0, y_0)$ we want to find such an edge that if the point belongs to any edge, the point lies on the edge we found, otherwise this edge must intersect the line $x = x_0$ at some unique point (x_0, y) where $y < y_0$ and this y is maximum among all such edges. The following image shows both cases.



We will solve this problem offline using the sweep line algorithm. Let's iterate over x -coordinates of query points and edges' endpoints in increasing order and keep a set of edges S_x . For each x -coordinate we will add some events beforehand.

The events will be of four types: *add*, *remove*, *vertical*, *get*. For each vertical edge (both endpoints have the same x -coordinate) we will add one *vertical* event for the corresponding x -coordinate. For every other edge we will add one *add* event for the minimum of x -coordinates